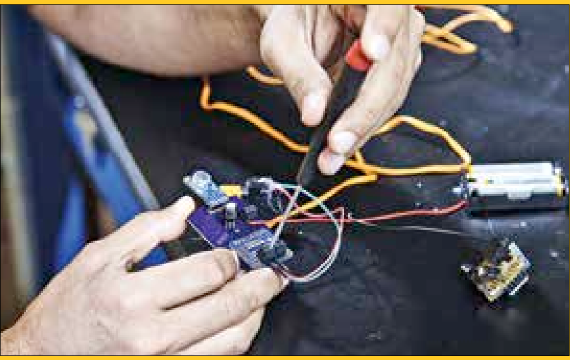
LoCHAid

Saad Bhamla, an undergrad from Mumbai, when looking to buy his grandparents hearing aids, found that the prices were absurdly high. 15 years later, now a bioengineer, he has invented a do-it-yourself hearing aid, made with inexpensive and easy-to-find materials. With this device, millions around the globe could finally restore their hearing, for less than a dollar.

As people grow older, a large percentage tend to develop hearing loss. High-pitch sounds, like beeps, speech sounds such as "s" and "th" for example become harder to distinguish. This can also lead to faster cognitive decline.



The solution? Hearing aids. However, custom hearing aids tuned to the needs of a user can cost as much as $5000 a pair! Non-customisable ones are still pretty pricey at $500.

To that end, Bhamla and his team set out to develop a cheap hearing aid built with off-the-shelf parts. They did this by soldering a microphone onto a small circuit board to capture nearby sound. Then they added an amplifier and frequency filter, to specifically increase the volume of high-pitch sounds to over 1000 hertz. They also added volume control, on/off switch, and an audio jack to plug in earphones. There's also a battery holder. Everything put together, LoCHAid is as small as a matchbox, and can be worn around the next. The device boosts the volume of high-pitch sounds by 15 decibels, while preserving volumes at lower pitches. It could also filter out loud sounds like dog barks and car horns. It also helped improve speech recognition. Again, all for under a dollar!

Airbus Zero E

Airbus has debuted three concepts for what would be the world's first "zero-emission" commercial aircraft. The concepts will supposedly come into service as soon as, or even before, 2035. The three new concepts are supposedly different from previous attempts to achieve the same, and are more experimental. The goal of course is to be able to provide long-distance air travel without producing greenhouse gases. If aircraft stop creating harmful carbon emissions, it would be a considerable boon to the Earth's atmosphere.

All three of the zero-emission concepts use hydrogen for fuel. As a result, the only emission is water vapor, which is fairly harmless. If the system works for aircraft, it becomes a viable and clean fuel alternative for other vehicles as well.

The first concept Airbus uses a turbofan design, with a modified gas-turbine engine fueled with hydrogen instead of jet fuel. The second concept uses the turboprop design and would use the modified gas engine for short-distance trips. The third concept, pictured above, involves a plane with a rather wide body, "blending" with the plane's wings, allowing for more room and better hydrogen storage options.

InvestEGGator

Sea turtle populations have been dropping fast, and they're on the verge of extinction. These animals usually lay their eggs on beaches, and the young hatchlings have to make the dangerous trip back to water without dying. However, the biggest threat to them aren't natural predators, but humans. They have been killed for their meat and shells by humans, who also destroy their nests and steal their eggs. To that end, conservation scientist Kim Williams-Guillen has come with a "golden egg". Or rather, a 3D-printed fake egg embedded with a GPS transmitter. The InvestEGGator as it's called, looks exactly like a sea turtle egg, and is meant to fool poachers who sell the eggs as food. Thanks to the GPS transmitter on the eggs, it's possible to track the thieves and even the buyers. Williams-Guillen even uses a pliable plastic material, called Ninjaflex, when 3D-printing the egg to give the InvestEGGator the feel of a real egg. Out of 101 eggs that were placed in four beaches around Costa Rica, 25 were taken by thieves.